Anti-derivatives of exponential functions

1

a $f(x) = \frac{1}{2}e^{2x} + C$

b $f(x) = -\frac{1}{5}e^{-5x} + C$

c $f(x) = 4e^{0.25x} + C$

d $f(x) = 3e^{3x} + C$

e $f(x) = e^{x} - \frac{e^{-3x}}{3} + C$

f $f(x) = \frac{e^{3x}}{3} + \frac{e^{-8x}}{8} + C$

g $f(x) = \frac{1}{3}e^{0.5x} + \frac{1}{18}e^{3x} + C$

h $f(x) = \frac{e^{2x}}{2} + \frac{2x^{\frac{3}{2}}}{3} + C$

2

a $-e^{-x} + C$

b $\frac{1}{2}e^{2x} + C$

c $2e^{\frac{1}{2}x} + C$

d $2e^{2x} + C$

e $-\frac{1}{3}e^{4-3x} + C$

f $-\frac{1}{2}e^{3-2x} + C$

g $-2e^{1-2x} + C$

h $2e^{0.5x} + \frac{e^{3x}}{3} + C$

i $e^{t} - 5t + C$

j $\frac{e^{4t}}{4} + \frac{t^{3}}{3} + C$

k $\frac{e^{3v}}{3} + \frac{v^{5}}{5} + C$

3

a $y' = x\left(2 + x\right)e^{x}$

b $x^{2}e^{x} + C$

Find value of C

4 $y = \frac{1}{3}e^{3x} - \frac{4}{3}$

5

a $2e^{4x} - 5x + C$

b $y = 2e^{4x} - 5x + 5$

6

a $-\frac{1}{2}e^{-2x} + 4x^{\frac{3}{2}} + C$

b $f\left(x\right) = -\frac{1}{2}e^{-2x} + 4x^{\frac{3}{2}} + \frac{1}{2}e^{-2} - 1$

7

a $3e^{x} - e^{-x} + C$

b $f\left(x\right) = 3e^{x} - e^{-x} + 2$

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8

a $y = \frac{1}{2}e^{2x} + \frac{11}{2}$

b $y = e^x - x^2 + 3$

9

a $k = 3$

b $y = \frac{e^{3x}}{3} + \frac{2e^3}{3}$

10 $y = 3e^x + 2x - 4$

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